



GM Data Insight

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2026-06-08

Manufacturing Operations Performance Analytics



Power BI Analytics Project

Business Problem

- Manufacturing companies generate large amounts of operational data from production processes, machines, and work orders. However, this data is often dispersed across multiple systems and reports, making it difficult for management to gain a clear view of operational performance. Without a centralized analytics solution, production managers face several challenges:
 - * Limited visibility into machine effectiveness and utilization
 - * Difficulty identifying production bottlenecks
 - * Inconsistent monitoring of key manufacturing KPIs
 - * Delayed detection of productivity issues
 - * Poor visibility into energy consumption and resource efficiency
 - * Time-consuming manual reporting processes

As a result, decision-making becomes reactive rather than proactive, reducing overall operational efficiency and limiting opportunities for continuous improvement.

Solution

- To address these challenges, a Power BI analytics solution was developed to consolidate manufacturing data into a single performance management dashboard. The solution provides:
 - * Standardized KPI monitoring
 - * Productivity and performance analysis
 - * Monitoring trend analysis
 - * Material and energy consumption tracking
 - * Executive-level operational reporting

By transforming raw production data into actionable insights, the dashboard enables management teams to identify improvement opportunities, monitor operational performance, and support data-driven decision making.

Expected Value

- * Monitor manufacturing performance through KPIs
- * Analyze machine utilization and effectiveness
- * Analyze teams strong points and points to improve
- * Track productivity and operational efficiency
- * Support data-driven decision making
- * Improve visibility across manufacturing operations

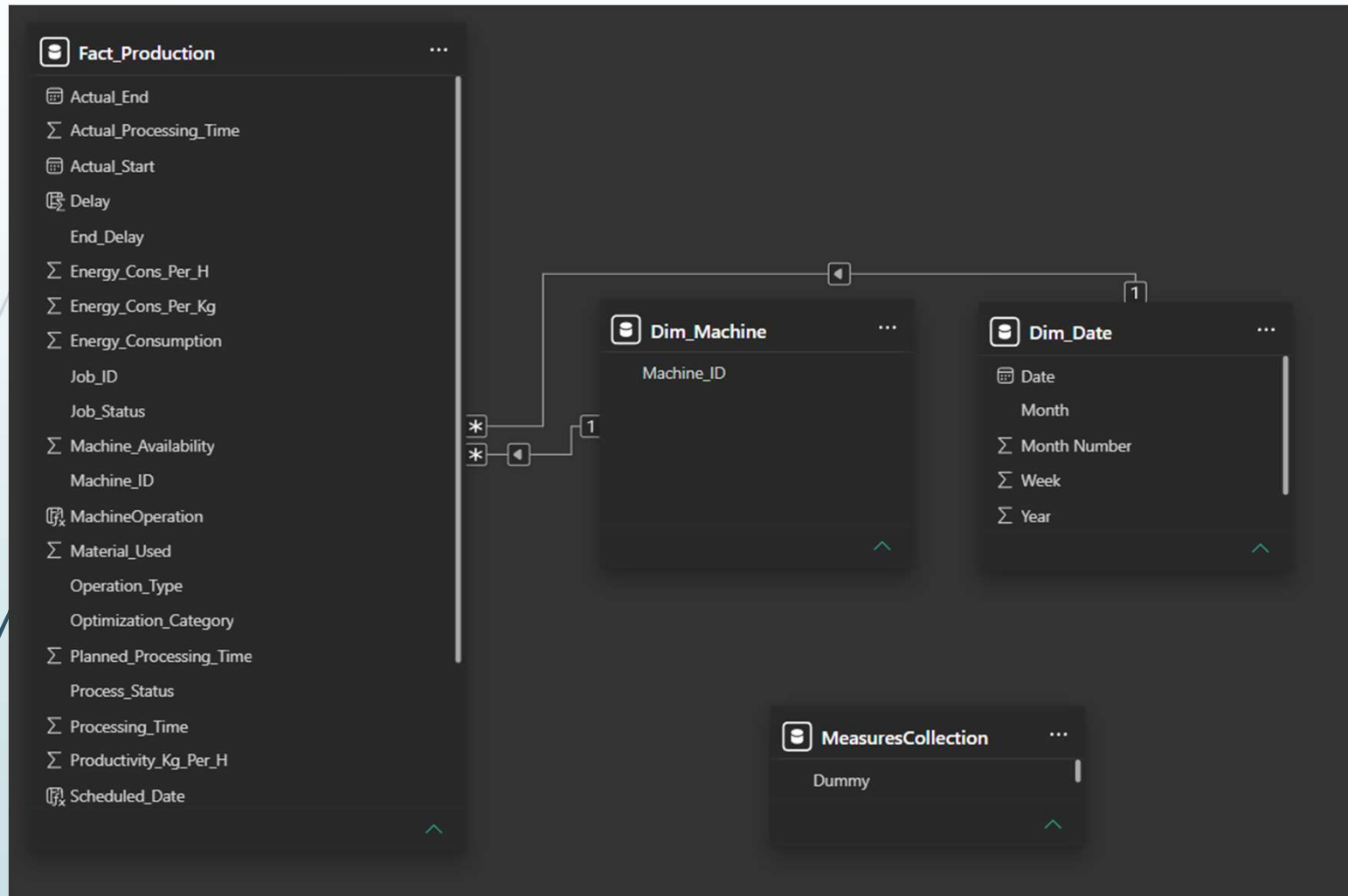
Source Data Set

Hybrid Manufacturing Dataset

Scope

Item	Value
Fact Tables	1
Dimension Tables	2
Dashboard Pages	4
KPIs	8+

Data Model

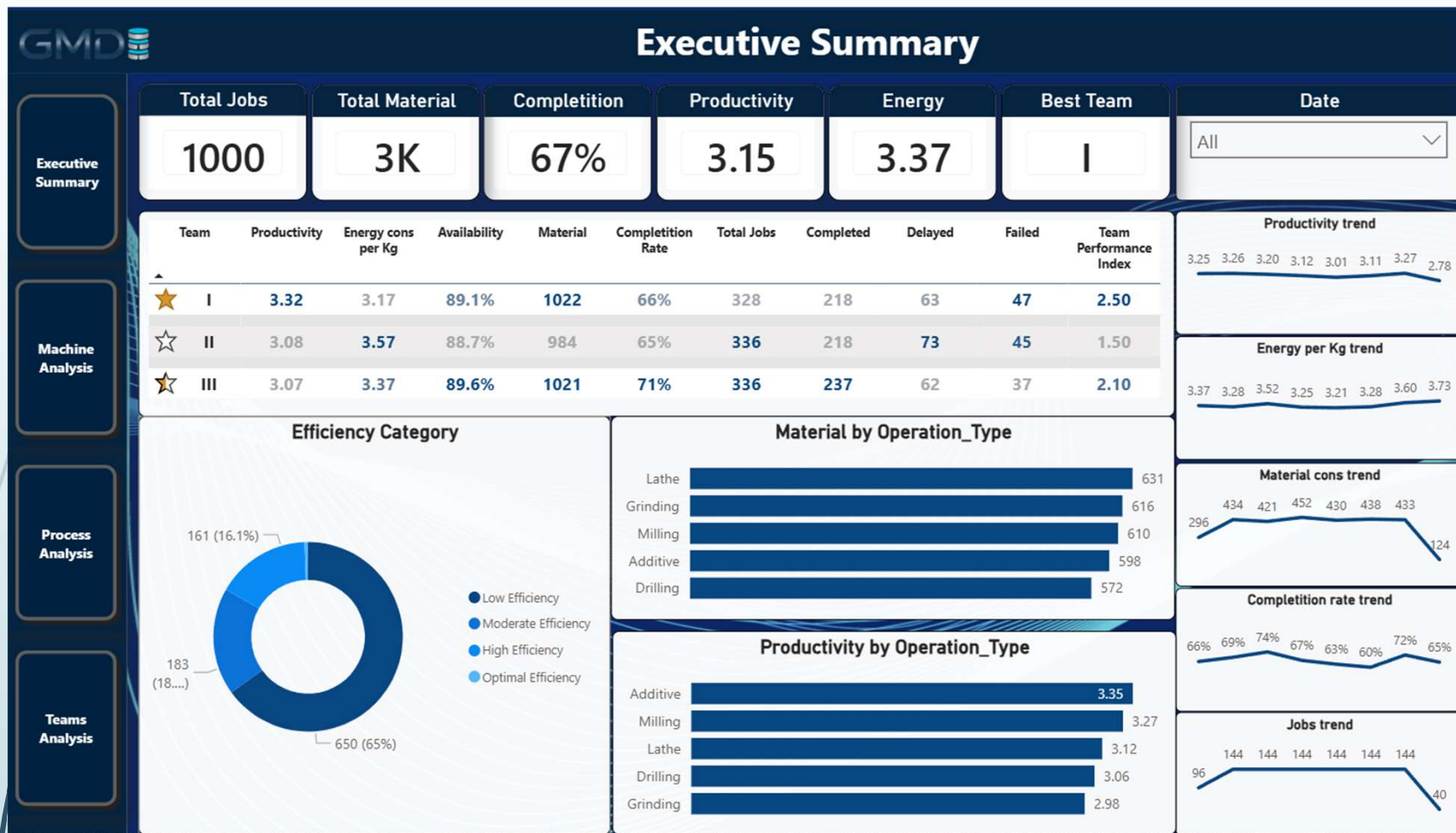


Star Schema
Single Direction Relationships
Fact + Dimensions

KPI Framework

KPI	Business Purpose
Productivity	Operational efficiency
Availability	Equipment utilization
Completion Rate	Execution quality
Energy per Kg	Cost efficiency
Total Jobs	Throughput
Total Material	Production volume

Dashboard Overview



Main insights supported

Productivity
Availability
Energy
Completion Rate
Team Comparison

Analysis Pages



Team Analysis

PURPOSE

The Teams Analysis page evaluates production team performance across key operational metrics.

MAIN KPIs

Productivity
Availability
Completion Rate
Energy per Kg

BUSINESS USE

This page supports workforce performance analysis, identifies best-performing teams, and highlights opportunities for productivity improvements

Machine analysis

PURPOSE

The Machine Analysis page focuses on machine-level operational performance and efficiency.

MAIN KPIs

Total Jobs
Availability
Average Productivity
Energy per Kg

BUSINESS USE

This page helps identify high-performing and underperforming machines, supporting capacity and planning optimization

Process Analysis

PURPOSE

The Process Analysis page provides detailed insights into operational performance across manufacturing processes

MAIN KPIs

Productivity
Availability
Completion Rate
Energy per Kg

BUSINESS USE

This page enables management to evaluate process effectiveness, identify operational bottlenecks, and optimize resource utilization across manufacturing operations.

Key findings

- 1. Team I Demonstrated the Highest Overall Performance
 - Team I achieved the highest productivity level while maintaining strong completion rates and competitive energy efficiency.
 - On the other hand, each team excelled in productivity on some process, and energy on some of machines.
 - This indicates that there is specific technical and organizational knowledge in each team which could be future benchmark.
- 2. Significant Productivity Differences Exist Across Manufacturing Processes
 - Additive process achieved the highest productivity levels, while Grinding showed lower productivity performance which dropped 4 days ago and still on low level.
 - This suggests opportunities for process optimization and further investigation into operational constraints affecting lower-performing processes.
- 3. Machine Performance Varies Considerably
 - Productivity analysis showed noticeable differences between machines, with some machine-process combinations significantly outperforming others.
 - In term of completion rate, there is indication of technical issues with some of the machines.
 - This variation indicates opportunities to standardize operating procedures and improve machine utilization and maintenance.
- 4. Energy Efficiency Differs Across Teams and Processes
 - Energy consumption per kilogram varied between teams and manufacturing processes.
 - These differences suggest that operational practices, machine utilization, and process characteristics have a direct impact on energy efficiency.
- 5. Completion Rates Indicate Opportunities for Operational Improvement
 - While the overall completion rate remained above 70%, delayed and failed jobs continue to impact operational performance.
 - Reducing delays and improving job execution could significantly increase overall manufacturing efficiency.
- 6. Process-Level Trend Analysis Revealed Performance Fluctuations
 - Several manufacturing processes showed fluctuations in productivity and energy consumption over time.
 - Continuous monitoring of these trends can help identify emerging operational issues before they impact overall performance.
- 7. Operational Performance Benefits from Multi-Dimensional Analysis
 - Combining machine, team, and process perspectives provided a more comprehensive understanding of manufacturing performance than any single metric alone. This demonstrates the value of an integrated manufacturing analytics approach.

Recommendations

➤ 1. Replicate Best Practices from High-Performing Teams

- Analyze workflows used by Team I.
- Examine operating practices of Team I on Milling and Drilling process.
- Examine operating practices of Team II on Additive process.
- Standardize successful operating practices.
- Share best practices across production teams.

Expected Benefit:

- Improved productivity and operational consistency across the workforce.

➤ 2. Investigate Low-Productivity Processes

- Compare process workflows between Additive and Grinding in term of productivity (speed, complexity).
- Evaluate machine setup and process execution procedures between Lathe and Additive.
- Identify gaps and go towards equalization as much as possible.
- Identify issue on 2023-03-22 on grinding process.
- Identify issue on 2023-03-23 on lathe process.
- Identify improvement on 2023-03-23 on Drilling process and standardize it.
- Identify workflow on 2023-03-24 on additive process and standardize it.

Expected Benefit:

- Higher throughput, improved resource utilization and achievable planification of production.

Recommendations

➤ 3. Investigate machine-process combinations performance

- Short term
- Plan Drilling process on M01 machine / avoid M02
- Plan Grinding process on M02 machine / avoid M04
- Plan Milling process on M03 machine
- Plan Additive process on M04 machine / avoid M01
- Plan lathe process on M05 machine / avoid M01
- Middle Term
- Standardize processes between machines according to best practice

Expected Benefit:

- Higher productivity and energy consumption performance.

➤ 4. Optimize Energy Efficiency

- Benchmark M02 regarding energy efficiency (spare parts, connections, speed, resistance)
- Benchmark Team2 efforts for energy performance in last 4 days.
- Implement energy optimization initiatives

Expected Benefit:

- Lower operational costs and improved sustainability performance.

Recommendations

- 5. Improve completion rates
 - Review planification of production.
 - Review and improve preparation and startup procedure for all process and equipment

Expected Benefit:

 - Lower operational costs and improved sustainability performance.

- 6. Track performance on daily basis
 - Organize daily routine for review and quick action for each KPI.
 - Inform production teams about results on daily basis

Expected Benefit:

 - Result oriented and competitive teams bring biggest impact on performance.

- 7. Expand Operational Performance Monitoring
 - Future versions could incorporate:
 - Downtime tracking
 - Root cause analysis
 - Predictive maintenance indicators
 - Quality performance metrics
 - Production forecasting

Expected Benefit:

 - Enhanced decision-making capabilities and greater operational visibility.

Technical Skills Demonstrated

Area	Skills
Power BI	Dashboard Development
DAX	KPI Measures
Power Query	Transformations
Data Modeling	Star Schema
Analytics	Operational Insights